

HR Attrition Analysis

Bangalore Office — Driver Analysis, Predictive Model, and Executive Summary

Business question:	Why is Bangalore's attrition higher, and can we predict flight risk?
Data:	204,000 employee records, cleaned to 201,000 after deduplication
Tools used:	PostgreSQL (cleaning, hypothesis testing), Python / scikit-learn (predictive model)

Executive Summary

Bangalore's attrition rate is 44.8%, compared to 27-31% across our other five offices. We tested five possible drivers using employee data, and confirmed all five as real contributors, ranked here by strength of effect:

Driver	Effect Size	Comparison
Job satisfaction	~13 pp	Low satisfaction: ~52% attrition vs High: ~39-40%
Overtime	~9 pp	Overtime: 50.9% vs No overtime: 41.9%
Salary hike %	~8 pp	Low hike (<6%): 51.4% vs Normal/High: 43.4%
Commute distance	~7.4 pp	Long (>25km): 51.3% vs Short: 43.9%
Tenure	~5.3 pp	Under 1 year: 49.3% vs 1+ years: 44.0%

Job satisfaction is the single strongest lever available. A predictive model built on these factors confirms this independently — satisfaction score accounts for more than double the predictive weight of any other single factor. Notably, even after accounting for overtime, satisfaction, commute, and salary hike together, Bangalore still shows elevated attrition risk beyond what these factors explain — indicating a location-specific issue not fully captured by current data (e.g. team-level management practices, local job market competition).

Recommendations

- Prioritize employee satisfaction initiatives in Bangalore first — this is the highest-leverage, most cost-effective intervention available based on both the driver analysis and the predictive model.
- Review and reduce mandatory overtime specifically in Bangalore, where nearly 30% of the company's total overtime hours are concentrated.
- Reassess the salary hike floor for employees currently receiving below 6% — this group shows meaningfully elevated attrition risk.
- Investigate what is specific to Bangalore beyond these five measured factors — the data shows a residual, unexplained gap that likely requires direct employee feedback (exit interviews, pulse surveys) to fully understand.

On predicting individual flight risk

A classification model was built to flag individual employees at risk of leaving. Using a balanced Random Forest model, it correctly identifies 58% of employees who go on to leave (recall), a substantial improvement over an initial, naive model that caught only 13%. This comes with a trade-off: roughly 57% of employees flagged as “at risk” will not actually leave (precision of ~43%). **This trade-off is appropriate given the low cost of a manager check-in versus the high cost of an unanticipated resignation** — the model is recommended as a screening tool to prioritize proactive conversations, not as a certainty about any individual.

Appendix A: Data Cleaning Procedure

1. Safe loading

All columns loaded as TEXT initially, regardless of true type, to prevent import failures from malformed values (e.g. ages like '17.0', mixed date formats) before any cleaning logic was applied.

2. Profiling before fixing

Checked total rows, duplicate rows, and null counts per column first. Found 204,000 total rows, 201,000 distinct rows (3,000 exact duplicates), and missing values concentrated in exit_date, age, education, satisfaction_score, commute_km, salary_hike_pct, and monthly_salary.

3. Duplicate removal

Removed 3,000 exact duplicate rows using SELECT DISTINCT, reducing the dataset to 201,000 rows.

4. Missingness investigation, not blanket handling

Confirmed exit_date nulls (135,105) exactly matched employees with attrited = 'No' — correctly understood as 'still employed,' not a data error requiring correction.

5. Location standardization

Merged 5+ spelling variants (Bangalore/bangalore/BANGALORE/Banglore/Bengaluru, and similar for other cities) into 6 clean city names using CASE WHEN with TRIM. This was the highest-priority cleaning step, since the entire business question depends on accurate location-level comparison.

6. Core hypothesis validation

Calculated attrition rate by cleaned location immediately after this fix, confirming Bangalore's 44.8% rate was genuine and not a spelling-driven artifact, before proceeding to further cleaning.

7. Age correction

Converted text values to NUMERIC (required due to decimal-formatted strings like '17.0'), then set biologically impossible values (≤ 18 , ≥ 80) to NULL rather than imputing a guessed value. This resulted in 150,682 unusable age values (~75% of the dataset) — age was excluded from the predictive model as a result, and this limitation is stated explicitly rather than hidden.

8. Department and gender standardization

Standardized casing and whitespace inconsistencies (including a double-space variant of 'Customer Support') using TRIM, REGEXP_REPLACE, and case-insensitive matching, with output verified after each fix.

9. Date parsing

join_date and exit_date contained three mixed formats (ISO, DD-MM-YYYY, MM/DD/YYYY). Each row's format was detected via pattern matching before parsing, with genuinely ambiguous two-digit/two-digit dates resolved using a documented default assumption (day-first for dashes, month-first for slashes), consistent with the approach used across all prior cleaning work in this portfolio.

Appendix B: Predictive Model Details

Approach

Features used: satisfaction_score, commute_km, salary_hike_pct, overtime_flag, and location (one-hot encoded, Bangalore as baseline). Age was excluded due to insufficient data quality (75% missing/invalid). Rows with remaining missing values in the selected features (approx. 3-5% per column) were dropped rather than imputed, retaining 177,788 usable rows.

Model comparison

Model	Recall (Leavers)	Precision (Leavers)	Accuracy
Logistic Regression (default)	13%	55%	67%
Logistic Regression (balanced)	56%	43%	60%
Random Forest (balanced)	58%	43%	60%

Why the default model was rejected

The default logistic regression achieved the highest raw accuracy (67%) but recall of only 13% on actual leavers — barely better than a naive model that always predicts 'stay' (which would score ~66% accuracy by only exploiting class imbalance). Since the business goal is identifying flight risks before they resign, recall on the minority class is the metric that matters, not overall accuracy. Applying class_weight='balanced' corrected this, at the cost of precision — an acceptable trade-off given a false negative (missing a real flight risk) is more costly than a false positive (an unnecessary check-in).

Feature importance (Random Forest)

Feature	Relative Importance
Satisfaction score	0.321
Overtime	0.132
Salary hike %	0.131
Commute distance	0.121
Location (combined)	0.295

Satisfaction score alone accounts for more predictive weight than any other individual factor, reinforcing it as the top-priority intervention area.

Note: This dataset was constructed to realistically represent HR attrition patterns for the purpose of this analytics project. All figures, cleaning decisions, and model results in this document are derived directly from that dataset using the methodology described above.